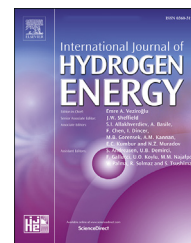


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Review Article

Prospects for scramjet engines in reusable launch applications: A review

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HIGHLIGHTS

- Requirements in reusable launch vehicle technologies are reviewed.
- Capabilities of scramjets to meet reusable launch system requirements are assessed.
- With hydrogen fuel, scramjet can enable sustainable and cost-effective RLV operation.
- Scramjet-RLV combine promises significant cost reduction in space missions.

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ABSTRACT

Partially or completely reusing launch vehicles significantly reduces expenses for manufacturing and testing in space missions. Thus, Reusable Launch Vehicles (RLV) can be a solution to achieving technological and financial goals associated with space programs. Scramjet engines have a higher specific impulse than rocket engines and the capability to operate efficiently at extremely high speeds. This paper reviews the requirements in reusable launch vehicle technologies and how the scramjet engines can be used with their advantages and limitations, and possible solutions to meet future requirements of reusable launch systems. The review covers different aspects of the launch requirements of a reusable launch vehicle, the applicability of scramjet in the RLV, and the key parameters to enhance the performance of the scramjet engine. With the possibility of hydrogen as fuel, it is inferred that the scramjet is a plausible candidate for the successful, sustainable, greener, and cost-effective operation of the RLV. The combination of Scramjet-RLV holds the promise of significantly bringing down the cost of space missions.

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Introduction

The space industry has been instrumental in the advancement of basic science, biology, physics, and mathematics. During recent decades, space research has offered promising solutions for disaster management and high-speed travel. Space tourism and interplanetary missions are fast emerging as niche segments of the space industry. Bringing down the expenses for space missions is a primary goal for the space sector around the world today. Typically, hardware accounts for about 80% of the total expenses in a space mission. Recovering and reusing the hardware for multiple launches might be a viable solution to decrease the huge cost associated with space missions [1]. It can also address orbital debris issues, which can cause hazards on Earth as well as in other space operations [2]. Greater launch flexibility and reusable operations are similar to aircraft making the scramjet technology one of the leading options for reusable launch applications.

A broad spectrum of concepts is available to completely reuse a single launch vehicle. It is required to compare these concepts with conventional launch vehicles to understand the feasibility of such applications. Tsuchiya and Mori [3] investigated the return of air-breathing launch stages after the separation of orbital stages of Mach 5.1 and 6.8 and found that the propellant requirement for return was about 5% of the initial propellant weight. Freeman et al. [4] addressed the technical and operational challenges of reusable launch vehicles for reducing costs from the earlier designs. Compared to air-breathing systems, rocket propulsion systems have a wide

range of applicability between a stationary state and orbit. But even the most efficient rocket engines have a specific impulse of only about 430s [5]. The efficiency of the rocket engines could be enhanced by implementing advanced rocket nozzles [6–13].

Characteristics of air-breathing engines are shown in Fig. 1 and Fig. 2. Among air-breathing engines, scramjet or supersonic combustion ramjet engine is a good candidate for hypersonic flight. Scramjet engines do not require a compressor, instead using shock wave compression in the inlet, allowing supersonic flow into the combustor. Combustion of fuel-air mixture happens at supersonic velocities [14]. The flow physics of supersonic combustion is complicated by the

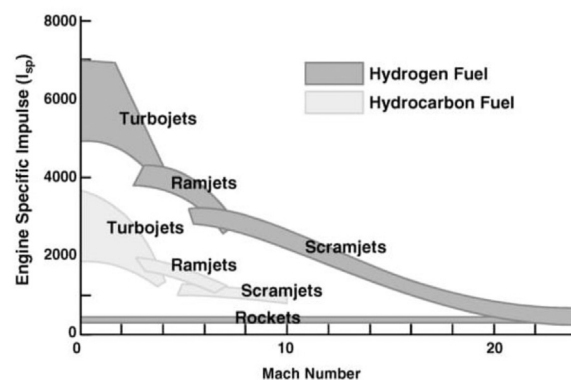


Fig. 1 – Characteristic performance for different types of engines [15].

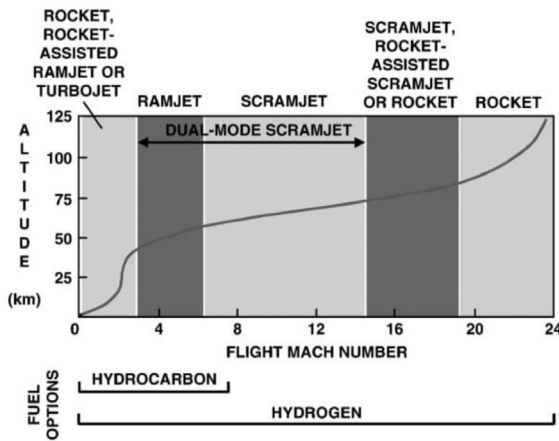


Fig. 2 – Engine options as a function of Mach number [15].

comparative rapidity of the combustion process. Supersonic combustion is sustained by maintaining the dynamic pressure above 9.5 kPa [15]. Applicability to launch vehicle systems can be improved by identifying and optimizing performance parameters to ensure reliable operation under complex flow phenomena. In the case of component performance, one of the critical components, the inlet isolator reduces the upstream back-feeding by increasing combustion chamber pressure. This is achieved by decoupling the inlet flow field from the combustion chamber. This will help prevent the inlet from unstart at a moderate flight Mach number [16]. When scramjet operates at Mach numbers greater than 8, the possibility of unstart diminishes [17,18].

Many issues need to be addressed for the successful application of scramjet engines in reusable launch vehicles. Some of the key issues regarding the intake, isolators, combustors, nozzles, trajectory, thermal management, and aircraft design have been addressed in this review. The cross-sectional schematic of a scramjet with different components is illustrated in Fig. 3. Different types of intakes for different configurations of scramjet have been discussed, such as axisymmetric and planar. Issues regarding unstart in the scramjet have also been briefly discussed. The isolator is a key component in separating the operation of the intake and the combustion chamber. Proper designing of the isolator can mitigate unstart at the intake. Various combustors of scramjet engines have been briefly discussed to improve the scramjet performance. The design used for scramjet nozzles for thrust improvement has also been reviewed.

Review methodology

This review has considered original works published in the past under the keywords of reusable launch vehicles, scramjet, and supersonic combustion, focusing on the physics which can be implemented in the execution of reusable launch vehicles. Google Scholar, Web of Science, and Science Direct database were used to collect the data.

The application of scramjet in Reusable Launch Vehicles (RLV) has emerged as an important focus area in aerospace engineering research. There are not many attempts at a comprehensive review on the studies. A majority literature trend was primarily on numerical simulations with limited experimentation. The topic is addressed as per the following steps: In the first part, the requirements of the relaunch or reusable launch application and current technologies involved in reusable launch vehicles are discussed. In the second part, the scope of supersonic combustion, especially the conventional scramjet concept, is investigated. In the third part, limitations are summarized, and ways to overcome these constraints are reviewed based on inferences from the literature is summarized in the fourth part.

Requirements in launch systems

Space missions that are complex and expensive in nature require highly reliable systems. To improve demonstrated reliability over the life cycle performed, independent mission assurance can have cost-effectiveness as well as reliability assurance to the mission [19]. Reusable launchers should provide cost-effective [20,21], reliable [22] sustainable substitutes taking into account the life cycle [23] and atmospheric impact [24], in aircraft design, manufacturing, and operations [25,26].

Payload requirements

Advancements in space technology and mission goals create requirements that have to be met by the launch systems. Conventional missions use large launch systems with multiple goals for long periods. The advancements in electronics facilitate small structures that can perform complex functions or may offer better performance with small satellites at lower costs. For example, systems involving multiple small satellites at lower altitudes for earth observation allow for cost reduction and improved performance compared to traditional satellites [27]. Small satellites can significantly bring down the costs involved in data acquisition for space weather research

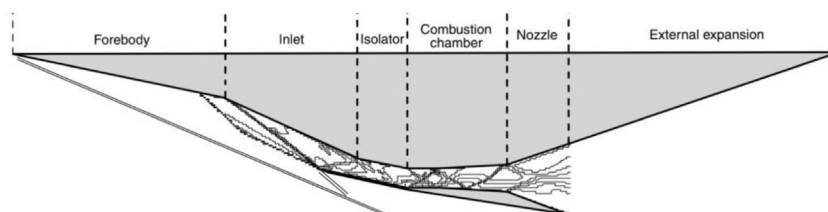


Fig. 3 – Schematic of a scramjet engine [14].

[28]. Since these satellites are required to be placed in a complex orbital schedule, they are sensitive to launch conditions [29]. Reusable launches can also facilitate more viable commercial passenger transport [30]. If structural redundancy and characterization of vibration environment are low for reusable vehicles, it is high for commercial aviation. Similarly, the severity of the aero-heating environment is high for reusable vehicles, while it is low for commercial aviation [19].

Launch frequency, manufacturability, and cost

The process of integrating technologies on a reusable launch vehicle by evaluating the size, and mission performance, and estimating the expenses for a conceptual design is an iterative, multidisciplinary effort [30]. The Operational cost for inspection, repair, and recertification of the vehicle between flights is significantly higher than the expendable launch vehicle's operational costs. Diminished payload capability due to the additional equipment for getting the vehicle back and in acceptable condition for reuse is another concern for reusable launch vehicles [31]. The flight of a reusable launch vehicle with the requirement for sophisticated controls to operate successfully is significantly more complicated than a flight of an expendable launch vehicle [32].

A thorough process validation is required to ensure safety standards. For reusable launch systems, this can be a configuration management process to track the flight and obtain details about its maintenance history and the remaining life of flight hardware for reuse [19]. The structure is expected to maintain its structural integrity during its designed service life when flown at design conditions [33]. In the case of reusable systems, components, and manufacturing processes should account for expected usage in different environments with several loads and stresses imposed on flight subsystems. To account for additional events, the qualification test program of reusable elements audits planned life cycles, such as flight, re-entry, recovery, transport, refurbishments, and ground operations [19].

Reliability

Reusability of the system is made possible through refurbishment activities such as inspection, testing, and replacement of consumables and non-conforming parts [34]. Three major domains of flight dynamics, subsystems, and flight systems need to be evaluated to understand the overall entry, descent, and landing of the system [35]. In order to address re-acceptance requirements for hardware that is substituted, an acceptance test program can be done. Local flight environment data per subsystem from sensors will reflect loads and stresses [19]. Structures standards like AIAA-S-081B5 (COPVs), AIAA-S-110-20,056 (Structures), and AIAA-S-080A7 (Pressure Components/Vessels/Structures) for National Security Space are some existing standards that can be applied to reusable systems. Considering larger factors of safety can account for additional margin coverage for the design for fatigue and fracture [19]. In addition to that, components such as propellants can influence tankage design and weight, insulation requirements, thermal fatigue cycling, and engine start and operating characteristics [36].

Several assessments are required to evaluate a launch vehicle's reliability. Critical components such as thrust chambers require identification concerning Low Cycle Fatigue (LCF) and High Cycle Fatigue (HCF) using various verification methods and implementing Fatigue Analysis Factor (FAF), including main combustion chambers subcomponents as well as turbomachine elements. Moreover, a Service Life Factor with a demonstration of the minimum calculated life of 4x service life for LCF and 10x service life for HCF must be shown for alternating and mean stress/strain to account for the effect of the stress concentration factor [37–40].

Environmental impact

The environmental impact has to be a major concern in all technology development processes. To make futuristic technologies sustainable, adverse environmental impacts should be minimal during the entire life cycle of the product. Without downplaying the contributions of fossil fuels in building the modern world, there is no denying the fact that the pollution resulting from fossil fuels has been triggering an environmental backlash affecting the quality of life on Earth [41]. Indirect exposure to chemicals involved in aircraft may come from soil or water, and direct exposure primarily occurs through atmospheric distribution, which poses environmental as well as health issues, and is toxic to humans and other living organisms. So, there is an urgent need for increased energy production from renewable and sustainable sources [42–44]. Hydrogen with good net calorific value and high flammability can be a good option [45].

Scramjet in launch applications

Scramjets operate in hypersonic conditions. Another system is required to put the scramjet in conditions where it can operate by itself. Hence, scramjets are used for launch systems as one component in multiple-stage systems. Some studies have discussed the application of scramjet in relaunch applications. Preller and Smart [46] described a three-stage rocket - scramjet - rocket system with a reusable second stage powered by scramjets. The system developed more than twice the average specific impulse of a rocket, as shown in Fig. 4. This model is reportedly capable of placing payloads up to 100 kg into a sun-synchronous orbit. Mehta and Bowles [47] mooted the concept of models having first-stage air-breathing systems, including a scramjet. A study comparing hydrogen and hydrocarbon-powered scramjet engines for orbital insertion observed significant improvement in payload performance if air-breathing propulsion is employed up to a flight Mach number of 7 [48].

Advantages of scramjet as a launch vehicle

Scramjet engines can generate high Mach numbers, up to 15, with a lightweight, simple geometry. Also, the scramjets carry only the fuel, using oxygen in the atmosphere for combustion [46]. This makes the vehicle lighter, smaller, faster and enables it to carry heavier payloads. Additional advantages in maintenance and manufacturing make it an apt technology

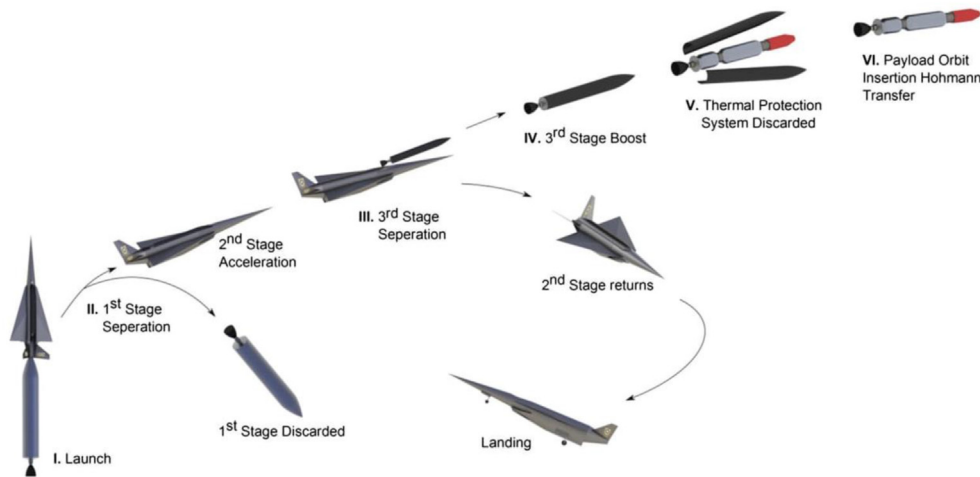


Fig. 4 – Scramjet in launch application [46].

for low-cost access to outer space with less complexity in engine design [49]. Scramjets also provide significantly higher specific impulses [50,51]. With the possibility of fully reusable aircraft-like operations, air-breathing propulsion has increased launch flexibility as a benefit for access to space. If properly executed, reusable systems will decrease costs, and aircraft-like operations help avoid the rigidity of fully rocket-based systems in terms of orbital inclination, time to rendezvous, and safe abort [52]. An air launch itself can have a lot of advantages. While penetrating the dense lower atmosphere, ground-launched rockets spend a disproportionate share of their energy. The performance of a ground-launched rocket capable of reaching 20 km can be extended to almost 100 km if it is launched from 20 km, and it would reach 60 km if it is launched from 11.5 km [53]. The drag can result in a loss of kinetic energy dissipated into displaced air, and some will go into self-heating [54]. Therefore, an air-launch reduces drag and heating since it is lifted above the lower atmosphere.

One of the main factors which affect scramjet engine efficiency is air-fuel mixing. Introducing cavities in the design geometry and injecting fuel inside these cavities for better mixing is one method for efficiency improvement [49]. Overall and propulsive efficiencies of an ideal scramjet engine are shown in Figs. 5 and 6. Changes in efficiencies with M_o (Mach number at freestream condition) are shown. Both efficiencies improve with increasing Mach number. An ideal scramjet engine operating at $M_o = 3$ has more efficiency compared to a ramjet engine operating at $M_o = 3$ [55]. Also, drag in the combustion chamber of a scramjet engine reduces its efficiency.

With an upstream injector for a small combustion chamber, the skin friction drag is reduced. Moreover, using barbotage injection with hydrogen/air and liquid kerosene helps in droplet breakage for efficient mixing with the supersonic air stream. Combustion can be sustained by injecting fuel in a series of pulses, and combustion never reaches a steady-state. Drag over a cavity is less when compared to a bluff body if a cavity flame holder type injection mechanism is employed. Since the shock is formed under injectors, cantilever fuel injectors provide a contiguous shock surface along the spanwise direction of the injector array. It will enhance the baroclinic effect, thereby providing greater mixing efficiency [56].

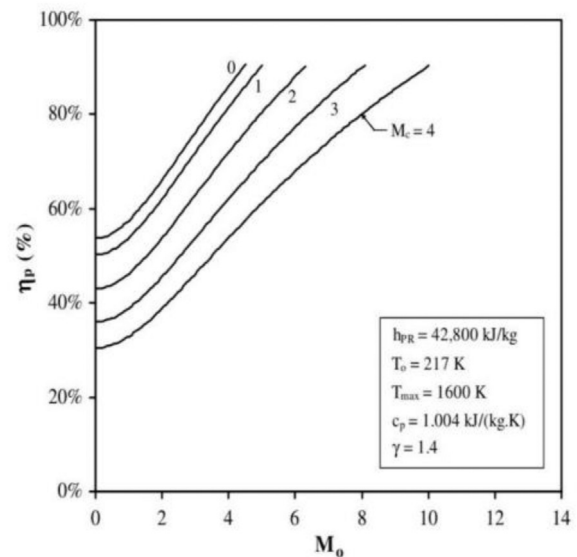


Fig. 5 – Ideal scramjet vs freestream Mach number: propulsive efficiency [55].

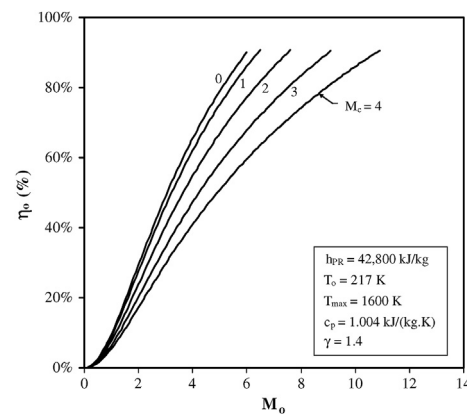


Fig. 6 – Ideal scramjet performance vs freestream Mach number: overall efficiency [55].

Limitations in using scramjet

Even though scramjet is considered an important air-breathing flight choice for the reusable launch stage, current scramjet technologies have certain limitations. To get into the orbital speeds, it is required to have fuels with high energy efficiencies along with good component efficiencies [57]. Scramjets operate under complex flows, which result in high temperatures and structural loads. It is necessary to ensure safe and efficient working under these operating conditions. Considering components of the system, for example, the intake of the scramjet, the performance varies with design parameters. Some design settings, like three ramp intakes, can cause considerable pressure loss and affect compression efficiency since ramp corners can cause shock waves [58]. Design, testing, and instrumentation costs are high, with large pressure loss and insufficient air-fuel mixing.

There are two constraints, known as air-breathing flight corridors, encompassed by constant dynamic pressure trajectories of 20 and 100 kPa [17]: first, the need to operate an air-breathing engine at a high altitude, and second, the structural limits of the vehicle at a lower altitude, either for cruise or ascent to low Earth orbit [14]. In launch applications, scramjets are operated in different altitudes, and their performance characteristics need to be evaluated. Too much compression in scramjet inlet can lead to reduced cycle efficiency as well as incomplete combustion of fuel and thus affecting performance. Moreover, high compression may lead to the requirement of bleed or variable geometry. Scramjet efficiency reaches the optimum level when the compression ratio is between 50 and 100 [14].

By optimizing the compression ratio at the inlet, fuel equivalence ratio, and water content, maximum specific thrust can be obtained. Maximum specific thrust indicates the isentropic efficiency of the inlet [59]. When the compression ratio at inlet and equivalence ratio is increased, the minimum free stream Mach number for a dual-mode scramjet also increases [60]. Viscous drag increased linearly with the dynamic pressure for a Mach number ranging from 7 to 10 and flight altitudes of 28–40 km [61]. At higher altitudes, the airflow density is reduced, and the scramjet engine thrust will fall off. In order to increase the final flight altitude and overcome the vehicle drag, the theoretical maximum specific thrust is estimated [62]. In the case of hypersonic vehicles, as scramjet-powered vehicle during ascent travels to high altitudes along an ascent trajectory, the static pressure in the scramjet combustor will decrease, which leads to engine flameout and lower combustion efficiency [63]. Duplication of freestream conditions is considered one of the main limitations of hypervelocity scramjet ground testing [64].

Hank et al. [65] pointed out that the number of payloads that can be put in orbit is limited by the launch cost, which is \$10,000 per pound at the time for low earth orbits. Payload capacity is reduced if hydrogen is used as fuel in scramjet applications [66]. The scramjet would have an active life of less than 20s [67]. At greater speeds and higher altitudes, engine efficiency is low on account of the short duration available for combustion. A method of supplementing the captured atmospheric air and increasing the efficiency of scramjets at

high altitudes can lead to better overall performance [68]. The large temperature differences between the outer wall of the cooling jacket and the inner wall attached to the engine structure result in dominant thermal stresses and strains. This can lead to LCF cracking and failure [69] of the REST scramjet, and its high compression inlets are not self-starting [70]. Li et al. considered fuel reactivity as a factor in controlling the self-starting of scramjet combustors [71]. Different combustion modes that exist in the scramjet combustor concerning flight Mach number, flight altitude, or fuel amount are another challenge faced by scramjet technology [72].

Overcoming limitations in applicability

There are several optimization studies to improve the applicability of scramjets. Different parts of the scramjet, namely the inlet, isolator, combustor, and nozzle, have been individually evaluated by the researchers to improve the performance of the scramjet. Other than the different sections of the scramjet, optimization of trajectory, thermal management, and aircraft design also plays a key role in performance enhancement.

Intakes

Apart from converting air to desirable conditions, intake plays an important role in performance since intake drag and compression efficiency depends on it [73]. Researchers exploring Busemann scramjet intakes conducted various studies [58,73–78]. Fig. 7 shows a schematic diagram of a scramjet having an axisymmetric configuration.

A study on Busemann-based intakes has suggested key design factors in which scramjets can be operated efficiently and robustly over their operating conditions, such as free-stream conditions on an ascent trajectory [58]. In comparison with configurations such as two-dimensional planar and sidewall compression, the axisymmetric configuration is more efficient, although it has inherent difficulty in starting spontaneously during flight [78]. Methods such as instantaneous diaphragm rupture and bleed slots [79], and streamline tracing [80] are suggested to give self-starting ability. While evaluating axisymmetric intake performance, it can be seen that it produces higher compression efficiency with a high uniform exit flow [78]. Krishnan et al. [81] conducted a numerical study on the effect of Shock-Wave/Boundary-Layer Interactions (SWBLI) on a planar scramjet intake for a flight of Mach 8 using large eddy simulation (LES) that explained the role of the boundary layer on the laminar-to-turbulent transition. Kane et al. [82] investigated the effect of SWBLI

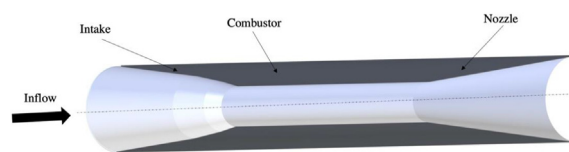


Fig. 7 – Schematic diagram of scramjet with an axisymmetric configuration [73].

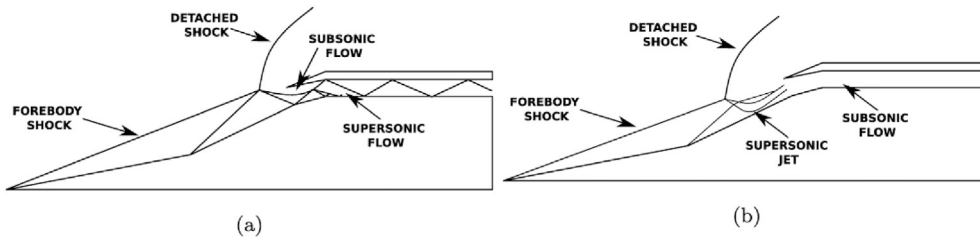


Fig. 8 – Schematic of (a) local unstart (b) hard unstart of a scramjet intake along with flow features [88].

on skin friction coefficient, performance, and surface pressure of intakes. A significant impact of Mach number on intake performance is demonstrated. Hohn and Gülhan [83] conducted an experiment on a scramjet inlet having a moveable cowl for changing the internal contraction. The experiment was conducted at Mach 7, and it was observed that the inlet performance improved and stabilized with the increase in internal contraction. Further investigations were conducted for Mach 6 and 7 by Flock and Gülhan [84,85] on straight and V-shaped cowl geometry. For increased yaw angles, the straight cowl had additional spillage of mass in comparison to the V-shaped cowl, which enhanced the intake starting. It was found that there was no unstart in the case of an internal contract ratio of 2.56 at the tested Mach 6 and 7 conditions. The v-shaped cowl has a slightly better mass capture ratio than the straight cowl geometry. An experimental study on the side clearance of the rectangular three-ramp scramjet intake was performed by Ogura et al. [86]. It was found that the side clearance of the third ramp expanded the starting range in comparison to the side clearance of the second ramp, which deteriorated the performance. It was also observed that with only side clearance of the third ramp, the mass capture ratio was enhanced, and a wider start condition was maintained. Devaraj et al. [87,88] conducted experimental and numerical studies on the unstart phenomena due to subsonic spillage near the cowl lip. It was noted that at the entrance of the isolator, the mean pressure reaches close to the unstart condition when the shock from the cowl arising due to the separation interacts with the shock due to reattachment at the isolator inlet. It was also observed that the shear layer originating from the shock triple point impinging on the cowl surface causes a self-sustained oscillation that could be seen at the local unstart. Fig. 8 shows the local and hard unstart of a scramjet intake. Sekar et al. [89] researched on the throttling dynamics of a hypersonic inlet by varying exit area of the isolator. An extended study was conducted by James et al. [90] to understand the separation bubble dynamics and the buzz characteristics. A detailed review of the unstart phenomena on the scramjet inlet-isolators has been conducted by Im and Do [91].

Isolator

Isolators are an important part of the scramjet that separates the rise in pressure due to the combustion in the combustion chamber from affecting the operations of the inlet [92]. Flow inside a scramjet isolator at different divergence angles is shown in Fig. 9.

Researchers have extensively examined the flow characteristics within an isolator [93–96]. Flow in the isolator is affected by both combustion and fuel penetration [97]. The range of equivalence ratio at which the scramjet combustion takes place improves with the help of an isolator [98]. Valdivia et al. [99] used a closed-loop control system with a combination of Wheeler doublet and vortex generator to mitigate and control the supersonic inlet unstarts. Shock train location inside the isolator has a significant impact on the isolator performance; therefore, Sethuraman et al. [100] used the suction method to control the shock train in the scramjet isolator. Shock train length was shown to have reduced with the help of suction, and the total pressure loss was improved. Axisymmetric scramjet isolator has better pressure recovery in comparison to planar scramjet isolators [101]. Researchers [102,103] have also tried to analyze axisymmetric scramjet isolators that are more prone to unstart in comparison to planar scramjet isolators. As the flow inside an isolator is complex in nature, Kong et al. [104,105] proposed a data-driven model for predicting the flow field inside a scramjet isolator.

Combustor

Efficient mixing and combustion within the combustor are crucial for the optimal performance of scramjet engines. Different types of combustors have been analyzed by researchers based on fuel injection and flame stabilization [106–121]. Abdulrahman et al. [122] conducted a review based on the subsonic and supersonic combustors used in aircraft. Two fuel injection techniques and variants are prominently explored by researchers, namely, cavity-based and strut-based fuel injection/stabilization [123]. In cavity-based fuel injection/stabilization, fuel is injected in the cavity or upstream of the cavity, and the flame is stabilized through the cavity [124–138], as shown in Fig. 10.

Choubey et al. [139] conducted a brief review of current advancements in cavity-based scramjet combustors. In strut-based fuel injection/stabilization, fuel is injected from the strut, which enhances mixing in the wake of the strut [114,115,140–151]. Different numbers and types of struts have been analyzed by researchers [152–166]. Fig. 11 shows a lobed strut configuration for enhanced turbulent mixing of the fuel [152,165,166]. Passive techniques have been explored by researchers to enhance mixing and combustion efficiency by modifying strut walls [167,168] and placing passive struts downstream of the strut-based fuel injectors by inducing shock and backpressure [169–172]. Different combination of strut, wall and cavity-based fuel injection has been

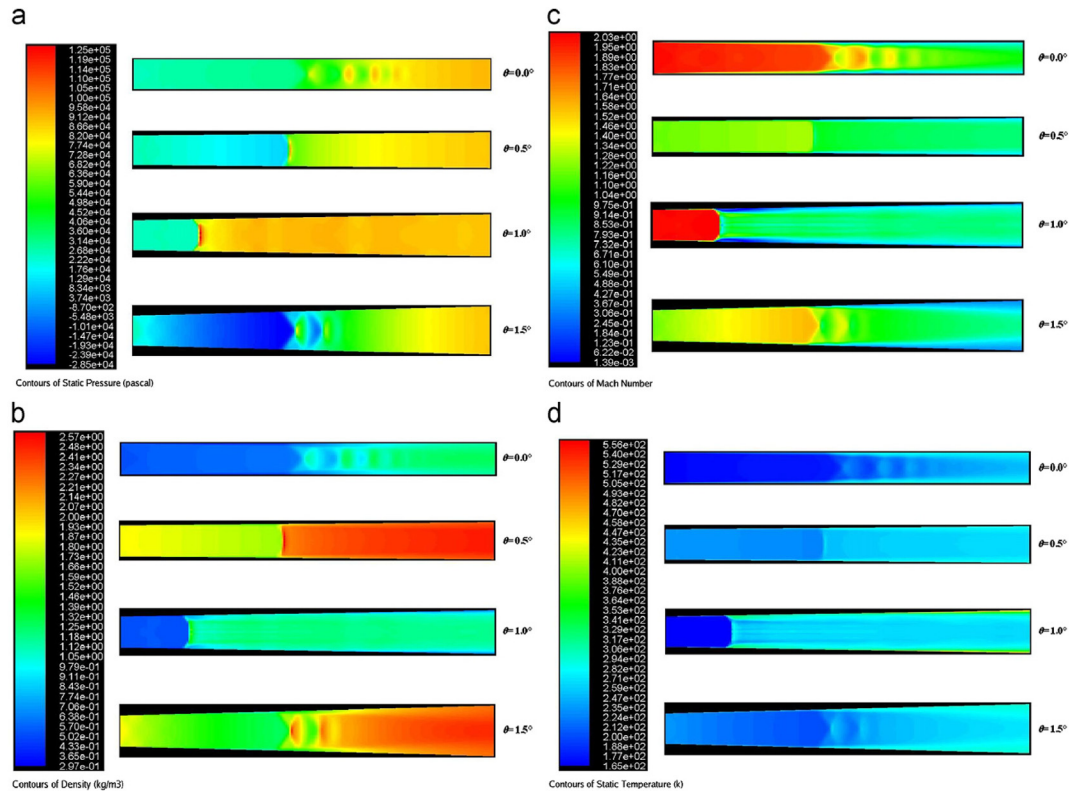


Fig. 9 – Numerical contour of static pressure, density, Mach number, and static temperature of scramjet isolators with different divergent angles [92].

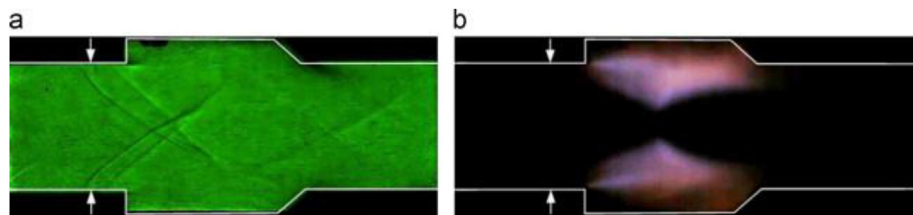


Fig. 10 – Experimental (a) schlieren image and (b) flame luminosity of a dual cavity scramjet combustor with upstream fuel injection [126].

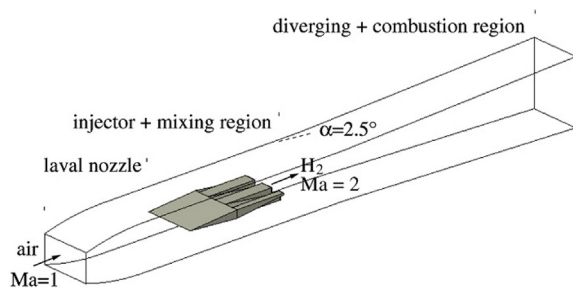


Fig. 11 – Geometry showing the scramjet combustor model having lobed strut fuel injector fuel [152].

implemented to enhance the combustion efficiency of scramjet combustor [173–178]. Annular-type combustion chamber [179–183] have also been explored by researchers, which has an advantage over rectangular geometries as it

doesn't have corner boundary-layer effects. The equivalence ratio also has a significant effect on enhancing the mixing and combustion [184–186].

Nozzle

Nozzles are an integral part of a scramjet engine that can augment the thrust with an optimal design. Bricalli et al. [187] conducted a study on the swept nozzle for Mach numbers ranging from 7 to 12. It was noted that the non-uniform compression had a higher specific impulse in comparison to uniform compression. Numerical studies have been conducted on the inverse design method for the scramjet nozzle [188–190]. There have been more studies [191–194], both numerical and experimental, on single expansion ramp nozzle (SERN) for scramjet (Fig. 12).

Li et al. [195] designed a scramjet nozzle using Rao's theory. Kunze et al. [196] proposed a method of design for a shape

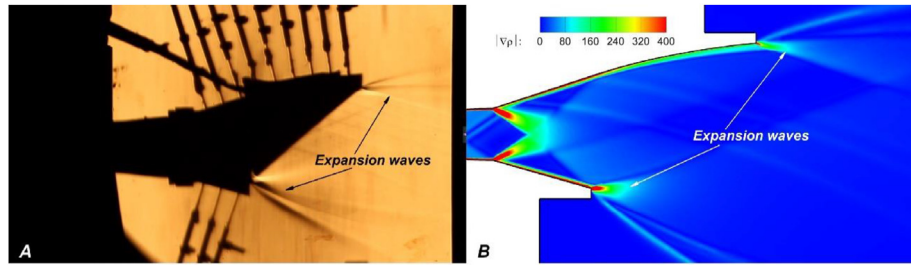


Fig. 12 – Flow field inside single expansion ramp nozzle (SERN) at NPR 103, (a) schlieren image, and (b) numerical density gradient [191].

transition scramjet nozzle that was applied to a nozzle having design Mach number 12. Cardon and Ogawa [197] analyzed strut-based fuel injectors with afterburners in an axisymmetric nozzle to optimize maximum thrust. For the RLV, improvement of the rocket nozzle is also a critical factor in the efficient operation. Researchers have been studying ways to use advanced rocket nozzles to improve the performance and efficiency of a rocket engine. These nozzles, also known as altitude compensation nozzles, can work effectively at a wide range of altitudes. Studies have focused on different types of advanced rocket nozzles [6], including dual-bell and double divergent nozzles [198–200], plug nozzles [201,202], expansion deflection nozzles [203,204], and dual throat nozzles [205,206]. These nozzles could be utilized for missions with reusable launch vehicles or single-stage-to-orbit missions.

Trajectory optimization

At higher Mach numbers, hypersonic air-breathing engines have lower efficiency [46]. Since the thrust produced depends on inlet conditions and dynamic pressure, the angle of attack and trajectory path is significant. Therefore, trajectory optimizations are required to get an efficient flight path. A study has determined the maximum payload to orbit trajectory, and the sensitivity of the trajectory is tested by varying the maximum allowable dynamic pressure and the drag produced by the vehicle [207]. This is also required on a return flight. Due to the large down-range distance travelled with a higher initial velocity at the onset of the fly-back trajectory, the orientation of the vehicle changes. Due to this, a full glide-back without fuel is difficult. For a fuel-efficient return, optimization techniques are thus required to be performed [208].

Thermal management

One of the challenges in high-speed flows is the short time available for the complete reaction. To augment the complete burning of fuels in supersonic flow, several methods can be employed [17,209]. Shock interactions can generate hot, high-pressure gas for combustion, and it can be beneficial for ignition [210]. In addition, the interaction of shock with injected fuel plumes will enhance mixing [211,212]. An experimental study on a non-intrusive method on low contraction ratio engines called “Thermal Compression” presented the first experimental evidence of thermal compression directly increasing combustion-induced pressure rise in a scramjet engine [213]. Various challenges in the combustion

process have to be resolved, including low residence time for the fuel droplets at supersonic speeds. Techniques for better mixing and reduction in pressure losses can augment combustion efficiency [214].

To tackle the complexities such as adverse combustor-inlet interactions, the selection of combustor geometries should be in such a way that it lends itself to structured engineering approaches with precise anchor and special control [215]. Plasma-assisted combustion, wherein plasma facilitates facile ignition of the air-fuel mixture, is another method for improving combustion. Quasi-DC discharge plasma ensures enhancement of combustion, which results in flame stabilization [216]. Another way is the two-strut scramjet engine which enhances the combustion and mixing efficiency through parameters such as strut tip diameter, divergence angle, and half-angle of a strut. For example, the efficiency decreases with the increase in strut tip diameter [217]. Zheng et al. [218] evaluated the possibility of film cooling in an isolator. It was observed that at optimal coolant flow conditions, the friction drag was reduced and had an enhanced backpressure resistance with reduced wall temperature. Yan et al. [219] conducted a regenerative cooling study on pylon-based fuel injection. It was found that the temperature of the pylon could be brought below the temperature limit of the material. Wei et al. [220] conducted LES study on the film cooling technique on the wall of a scramjet engine. It was found to have an additional reduction in skin friction and increased anisotropy of turbulence. Another study of thermal management on strut-based fuel injectors conducted by Li et al. [221] found that the strut can be cooled using a miniature-sized coiled cooling configuration inside the strut, which can be operated under a low mass flow rate, as shown in Fig. 13.

Design of the vehicle

Design methods such as profile variations can also influence performance. A study on three-dimensional shape transition scramjet has shown that an added contoured centre body can establish a maximum thrust nozzle. This allows for the adaption of nozzle shape to different geometries, eliminating usual drag-producing surfaces [222]. For commercially reasonable payloads, considering horizontal take-off single-stage-to-orbit, it gets bigger, although take-off mass is reduced by in-flight filling of oxygen from the atmosphere [223]. With kerosene for the initial stages of launcher applications, requirements for heavy storage tanks can be avoided

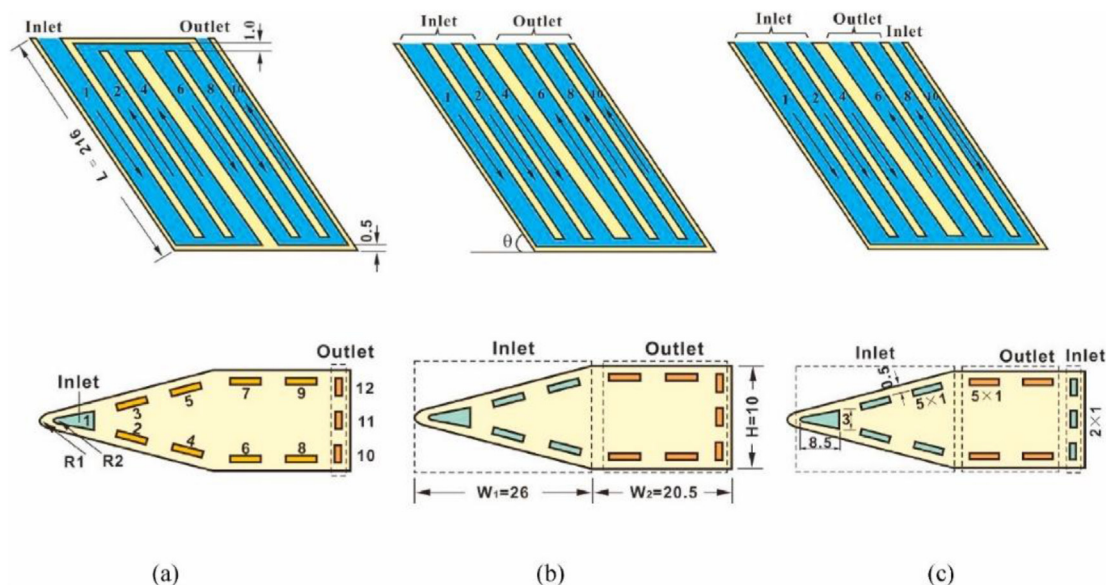


Fig. 13 – Miniature-sized coiled cooling configuration inside three types of struts: cross-sectional and top view [221].

[224,225]. Thus, a large single-stage-to-orbit for commercial applications requires careful optimization of propellants along with advanced structure, whereas smaller ones for defense forces may benefit from the use of hydrocarbons for air-breather mode [67].

There are a lot of complex processes involved in the design. The design merits can be discussed with approaches like the vehicle's take-off gross weight or dry weight. The take-off gross weight drives the operational costs and life cycle costs, while dry weight drives the initial cost [226]. Determining performance requires multidisciplinary approaches, and designs tend to be complex in this aspect. Designs have to be resized according to the mission requirement, and the ideal process will consider complex multidiscipline interactions and nonlinear effects of changing propulsion and internal system size [226]. Therefore, for a reusable application, multiple missions might face design challenges since each mission is different.

Further analysis of existing vehicle designs in aerodynamics, propulsion, and control systems has shown the merits that are promising to improve the applicability of air-breathing launch vehicles [227]. For simple fuel supply and compactness, solid-fuel scramjets can provide the benefits of adjustable thrust and re-ignition, making it the simplest air-breathing engine for hypersonic speeds [228]. Limitations such as difficulty in thermal conversion could be a challenge, although studies have shown that the potential high energy of boron-based propellants is promising [229]. Enhancing mixing [230], flow path design and optimization [231], and lack of data [228] are the other issues to be addressed in solid-fuel designs. Small changes in components can improve performance and applicability. For example, an alternating wedge injector strut showed better results compared to a circular injector [232]. Finally, to tackle intense heat at high speeds, it should be made up of materials with good properties to resist this high temperature and thus endure the intense heat [49].

Hydrogen as an energy source

Hydrogen, with its environment-friendly nature and wide range of production processes, can be a very good energy source for spacecraft expecting a long life of operation. However, the low volumetric energy density and higher flammability range are challenges for hydrogen as a propulsion energy source. Although low-density results in weight savings, for achieving a sufficient range in transportation applications, it becomes a necessity to increase the density of hydrogen. High-pressure storage as compressed hydrogen at 70 MPa, or cooling to liquid is the possible solution for tackling this issue [45]. Hydrogen may also pose safety concerns and incur huge costs associated with storage and manufacturing. Compressed hydrogen and liquid hydrogen are considered as most suitable options for onboard storage concepts. However, both storage systems demand sophisticated storage tanks [233]. Boretti [234] briefly discussed hydrogen as an energy source in commercial hypersonic aircraft and the technical hurdles faced in the implementation.

Conclusion

As space exploration and space travel gets popular every day, it calls for safe and reliable space technologies that can offer cost-effective space missions. Reusable launch vehicles open up opportunities for passenger travel and space missions. Scramjet can offer better performance characteristics compared to conventional rockets. Many reusable vehicle programs have found scramjet as a viable option for space applications. In this paper, studies on the prospects and possibilities for employing scramjet for powering reusable launch vehicles are comprehensively reviewed. When discussing the launch requirements, advancements in electronics have made small satellites that can operate individually as well as

in constellations. Hence, recurring space missions have to be launched with high precision. Scramjets have the capability to achieve precise launches. Scramjet implementation as a stage could also bring the advantages like lightweight and the high specific impulse to improve the whole mission technologically and economically. Thus the combination of Scramjet-RLV holds the promise of significantly bringing down the cost of space missions. However, Scramjets have certain limitations, such as a lack of self-starting nature and other design challenges. Improving the individual component performance of the scramjet can also result in an efficient performance. Although many studies have been conducted, further research is required to improve the efficiency and performance of intakes, isolators, combustors, and nozzles. Since the operation is in a supersonic region, accurate and efficient thermal management is also required. The lack of flexibility, with a single design for multiple applications, may limit the re-launch applications. These limitations and design problems can be addressed and resolved with further research. If green hydrogen is used as fuel, the scramjet offers a sustainable, greener, and cost-effective operation of reusable launch vehicles.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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